

When Peer Agreement Hides a Reversed Ranking: Graph Conditions for a Learning-Augmented Prediction Interface

Abstract

The survey Q treats rankings and numerical scores as distinct prediction objects and requires a declared error and access cost before transferring a prediction guarantee to a decision system. The local manuscript P identifies Bradley–Terry scores from peer reports under shared-draw conditional independence. We examine a restricted violation of that premise: every item has an independent, common binary flip of constant positive mean. Peer agreements retain exactly P’s rank-one form, yet a four-vertex comparison path admits two observationally identical worlds with opposite endpoint orders. Within this restricted model, comparison graphs of diameter at most two preserve all pairwise orders; on trees this condition is exact. A nondegenerate triangle or four-cycle identifies the common flip rate and hence the numerical scores, whereas a recorded six-cycle retains two admissible rates and reverses a pair. A separate finite-sample bound describes P’s clean-model score interface and its query count, conditional on an unobservable misspecification allowance. None of these statements proves truthful reporting or downstream decision performance.

1 Question and source boundary

Q is the learning-augmented algorithms survey by Zhao et al. [4]; P is the supplied local manuscript on peer-derived contribution rankings [1]. P’s internal names “Q” and “P” denote other papers and do not name the pair studied here. Q distinguishes ordinal rankings from point predictions, notes that prediction quality may be only partly observable, and calls for an error metric, a transfer condition, and accounting for prediction queries. P proves a population interface: under its shared realised comparison and conditionally independent symmetric peer flips, a calibration triangle and overlap paths identify positive observer reliabilities; calibrated edge means then identify Bradley–Terry differences on a connected comparison graph, up to a common score translation. The agreement-calibration step is already present in single-coin crowdsourcing, including sparse worker interaction graphs [2]. We claim no new rank-one calibration method.

The contribution here is an audit of the object passed from P to a decision algorithm of Q’s kind. Its negative example is exact at the level of the entire observable report law. Its positive graph statements require a constant positive common-flip mean, and its sample bound concerns the clean interface or a separately bounded moment deviation.

2 Model and P's conditional interface

Let $G = (V, E)$ be a connected undirected comparison graph with arbitrarily oriented edges. For edge $e = (i, j)$, the intended comparison $X_e \in \{-1, +1\}$ has $\Pr(X_e = +1) = \sigma(c_i - c_j)$, where $\sigma(t) = (1 + \exp(-t))^{-1}$. Thus $\mathbb{E}X_e = \tanh((c_i - c_j)/2)$. A second graph records which observers jointly label the same realised comparison. Under P's clean premise, observer r reports $Y_{er} = X_e Z_{er}$, where the Z_{er} are conditionally independent symmetric flips with positive means a_r . Hence shared-item agreements and edge means obey

$$q_{rs} := \mathbb{E}[Y_{er}Y_{es}] = a_r a_s, \quad \mu_{er} := \mathbb{E}[Y_{er}] = a_r \tanh((c_i - c_j)/2).$$

A triangle of observed overlap links gives $a_r = \sqrt{q_{rs}q_{ru}/q_{su}}$; observed paths propagate this calibration. Dividing μ_{er} by a_r , applying $2 \operatorname{artanh}$, and integrating along G gives c up to translation.

3 A report-law ambiguity under a shared flip

Suppose each item has an additional independent sign B_e with a common mean $b = \mathbb{E}B_e \in (0, 1)$, and reports are $Y_{er} = X_e B_e Z_{er}$. Draws and peer flips are independent as specified in the peer notes. Conditional independence given the intended X_e now fails, but

$$q_{rs} = a_r a_s, \quad w_{ij} := \frac{\mathbb{E}Y_{er}}{a_r} = b \tanh((c_i - c_j)/2).$$

Thus a perfect agreement factorisation does not certify that the calibrated mean targets the intended comparison. P's identification theorem remains correct under its stated premise; its suggestion that correlated errors necessarily destroy the observable factorisation is too broad.

Proposition 1 (Four-vertex ambiguity). *On a comparison path with four vertices, there are a clean world satisfying P's premise and a constant-common-flip world with the same complete distribution of peer reports, although their endpoint score orders are opposite. Consequently, from reports alone, one of these worlds has endpoint-order error probability at least 1/2 for any estimator, at any sample size.*

Proof. Orient the path $1 \rightarrow 2 \rightarrow 3 \rightarrow 4$, and in the common-flip world take edge differences $(4, -3/2, -3/2)$, so $c_1 - c_4 = 1$. Set $b = 1/2$. Define

$$f_b(d) = 2 \operatorname{artanh}(b \tanh(d/2)).$$

In the clean world take latent comparisons $X'_e = X_e B_e$, peer flips Z_{er} , and path differences $f_b(d_e)$. A path has no cycle constraint, so these differences integrate to a Bradley–Terry score vector. The observed reports are $X'_e Z_{er}$ in both worlds, while

$$c'_1 - c'_4 = f_{1/2}(4) - 2f_{1/2}(3/2) \approx -0.265 < 0.$$

The two report distributions coincide and the true endpoint orders differ. The two error probabilities for a binary order decision therefore sum to at least one. $\square$

4 Which graph information survives?

The next statements concern only the constant positive common-flip model and population means. They do not test whether observed data satisfy that model.

Proposition 2 (Ordinal boundary). *For every connected comparison graph of diameter at most two, the calibrated edge means determine every pairwise score order for every admissible $b > 0$. For comparison trees with $0 < b < 1$, diameter at most two is also necessary for universal order preservation under the uncorrected edge transform f_b . This is an ordinal statement and supplies no numerical score guarantee.*

Proof. For a one-edge path, positive b preserves the edge sign. For two consistently oriented path edges with differences x, y , f_b is odd and strictly increasing, so $\text{sgn}(f_b(x) + f_b(y)) = \text{sgn}(x + y)$. Every vertex pair in a diameter-two graph has such a path. Conversely, a tree of diameter at least three contains a three-edge path. Put differences $(2t + 1, -t, -t)$ on it. Its true endpoint difference is 1, whereas as t grows the transformed sum tends to $-2 \operatorname{artanh}(b) < 0$. $\square$

Short cycles can identify the missing cardinal scale under this same model. For a consistently oriented triangle $(i, j), (j, k), (i, k)$, write $A = w_{ij}$, $B = w_{jk}$, and $C = w_{ik}$. If $ABC \neq 0$, the addition identity gives

$$C = \frac{A + B}{1 + AB/b^2}, \quad b^2 = \frac{ABC}{A + B - C}.$$

The positive admissible root identifies b ; then every edge difference is $2 \operatorname{artanh}(w_{ij}/b)$, and connectedness identifies scores up to translation. This is population self-calibration within the specified class.

For four consistently oriented cycle edges, let S_k be the elementary symmetric sum of degree k in their calibrated means. Cycle consistency reduces to $S_1 b^2 + S_3 = 0$. If $S_1 \neq 0$, the positive admissible root of $b^2 = -S_3/S_1$ is unique. Thus a triangle is sufficient but not necessary. A cycle alone is not sufficient in general: the recorded six-cycle with calibrated means $(-65, 60, -52, -23, 62, 16)/100$ has admissible rates approximately 0.739 and 0.854, and the corresponding integrated scores give opposite orders for one vertex pair.

5 A conditional error and label budget

A separate calculation prices P's clean interface. Choose a calibration triangle and overlap tree whose greatest depth from that triangle is D , and let L_H be the selected overlap links. Suppose $a_r \geq \alpha > 0$, $0 < \tau \leq 1/2$, $p_e \in [\tau, 1 - \tau]$, and every required agreement or edge-mean estimate is within δ of its clean-model value. Let L_G be the unweighted comparison Laplacian and set

$$u = \frac{2\delta}{\alpha^2}, \quad T = (D + 3/2)u, \quad h = \frac{\delta e^T}{\alpha} + e^T - 1.$$

If $\delta \leq \alpha^2/2$ and $h \leq \tau$, P's triangle and path calibration, edge logit inversion, and centred least-squares reconstruction obey

$$\|\hat{c} - c\|_2 \leq \sqrt{\frac{|E|}{\lambda_2(L_G)}} \frac{h}{\tau(1 - \tau/2)}.$$

Here u bounds each log-agreement error, T bounds propagated log-reliability error, h bounds calibrated edge-mean error, and the Laplacian factor converts edge error to centred score error. This is an upper bound, not a matching rate or a decision theorem.

On every measured overlap link both observers label the same realised draw. With N independent items per measured population, $M = |L_H| + |E|$ measured moments, and population misspecification bias at most β , the same premise holds with probability at least $1 - \zeta$ for

$$\delta = \beta + \sqrt{\frac{2 \log(2M/\zeta)}{N}}.$$

A direct plan uses at most $N(2|L_H| + |E|)$ peer labels. This count is not yet a monetary or objective cost. The bias β need not shrink with N , and the path ambiguity shows why report-only checks cannot set it to zero relative to the intended target.

6 Related work and interpretation

Ma et al. already establish label-free single-coin worker calibration and rank-one agreements from sparse interactions [2]. That step is prior work. Bradley–Terry diagnostics examine model failures and ranking effects [3]. Our targeted searches found no exact constant-common-flip graph statement; this does not establish novelty.

For Q’s interface, the diameter-two result could serve a guarantee based only on ordinal error. P’s potential uses numerical scores, so it requires separate sensitivity and query-cost arguments even after cycle calibration.

7 Limitations and open questions

Identification assumes an edge-constant positive flip mean independent of intended comparisons and stable positive reliabilities. Short-cycle formulae cannot test these premises; edge-dependent or unrestricted shared bias remains unresolved. Degenerate cycles and finite-sample cycle stability are outside the result. The work supplies no truthful reporting, minimal general-graph audit design, policy sensitivity, or empirical validation. A next step is a conditioned finite-sample cycle bound with label costs, then error transfer to one specified decision rule. These remain open questions.

References

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